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(54) **STIRRED FREEZE DRYING**

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Description

[0001] The invention relates to a method and device for drying solutions and solids containing liquid, in which a vessel 1 is applied having a downwardly conical shape along the wall 2 of which a mixing member 3 rotates, in which said mixing member 3 rotates with a small inter-space along said wall 2, which vessel, 1 is provided with a jacket 4 and is connected to a vacuum system 5.

[0002] From US-A-4.245.399 a device is known for drying products of several types, which can be particulate materials with different humidities, comprising substances which are completely moistened by, for instance, organic solvents and possibly having an adhesive nature, which has essentially the above characteristics.

[0003] With respect to the term essentially, it is remarked, that there is meant that the above vessel 1 is a cylindrical vessel with a downwardly conical end, inside of which along the entire inner-wall 2 a mixing member 3 is provided which rotates with a small clearance.

[0004] The inter-space or play ("clearance") is at most 5 mm (viz. column 2, line 26-30). The provision of the mixing member 3 which rotates with a small clearance serves to effectively obviate the caking of the material to be freeze dried on the inner-wall of the vessel. 1.

[0005] The drying device known from US-A-4.245.399 is provided with a supply of a drying gas and in view of the consideration that one was foremost thinking about adhesive substances, not seen as a device for freeze drying relatively large batches of products.

[0006] Freeze drying or lyophilization is on the contrary the sublimation of a solvent that is crystallized at low temperature. The solvent is thereby sublimated directly from the solid phase into the vapour phase. By freeze drying a product can be obtained which does not perish easily and which will remain unaltered after combining again with the solvent. In other words, by the freeze drying process the material to be dried is frozen in its original structure and the solvent is removed from this structure without changing it. In the case for instance of an organic cell, the outer cellwall will remain its original shape at freeze drying, whereas at thermal drying the cell will slowly shrink and after moistening again the original structure will not completely be regained. Another important aspect of freeze drying is the low drying temperature, thus there is no thermal loading of the material to be dried.

[0007] The usual freeze dryers are of the "trays-in-cabinet" type. In that case the freeze dryer is a stationary chamber having a fixed number of trays (or plates) which are covered by the frozen substrate. Below the trays or plates a heating circuit is provided for supplying the energy for sublimation. After the closing of the chamber the system is evacuated and the sublimation process starts.

[0008] A great disadvantage of this type of freeze dryer is lump formation when one dries the material on a large scale. Notwithstanding the optimal structure of the separate product particles, the material on the trays or plates usually forms a hard cake of material. After freeze drying

the product must quite often be shattered or crushed, which can lead to damage of the product structure.

[0009] Another disadvantage is the relatively low heat exchange rate because of the resting condition of the material.

[0010] US 4,608,704 discloses a method and a device according to the preamble of claim 1.

[0011] The invention provides a method and a device according to claims 1 and 21.

[0012] Surprisingly it has now been shown, that by the application of a vessel having a downwardly conical shape of which the inside wall has a small clearance with the rotating mixing member, the disadvantages of the usual freeze drying process can be obviated.

[0013] With a so-called "stirred freeze dryer" according to the invention, which is operated at low temperature and low pressure, it appeared to be possible to obtain a lump-free and free-flowing product.

[0014] Stirred freeze dryers also have the advantage of a better heat exchange rate, because of the continuous mixing of the product to be freeze dried, which shortens the drying process.

[0015] Finally the application of a stirred freeze dryer often leads to a simplification of the freezing step, as this one can be made in the same device as that of the drying step. It is now unnecessary to have a separate freezing device. At the usual devices the material to be freeze dried is often frozen in separate freezing devices and thereafter loaded into separate vacuum drying devices.

[0016] Very often said conical vessel will form a part of a conical mixer, because of the good technical results obtained with conical mixers.

[0017] It is remarked, that cylindrical vessels, comprising a conical and having inside a rotating mixing member which has a small clearance with the inner wall of the vessel, are also applicable and can be used because of the advantage of space saving.

[0018] In connection with a good mixing in that case one has to take into account a certain dimensional ratio of the cylindrical part and the conical part. It is also remarked, that a mixing vessel of a so-called "classical" conical mixer often comprises near the lid a short cylindrical part, in connection with space of auxiliary machinery, such as drive engines and material supplies. Furthermore there is pointed to JP-2001263942 (5 pages), in which a conical mixer-dryer is described having a cylindrical part which is connected to the conical end; see figures on pages 1 and 4.

[0019] By the term conical mixer is also meant a so-called a double conical mixer, although the construction thereof is more complicated.

[0020] According to a variation of the novel freeze drying method, a batch to be freeze dried in the vessel is frozen by direct cooling.

[0021] A suitable way of direct cooling is loading liquid nitrogen or solid carbonic acid in the vessel and mixing it with the batch to be freeze dried.

[0022] Of course the cooling for the freezing of the

batch in the vessel can also take place by indirect cooling through the wall of the vessel by loading a suitable cooling medium in the jacket. As cooling medium a low temperature cooling medium is used.

[0023] From experiments it has been shown that it is favourable to freeze the batch to be freeze dried with a freezing rate of 0,1 - 10°C/minute. Of course the freezing takes place under stirring or movement of the mixing member, by which the freezing rate is controlled by dosing the direct cooling medium or the control of the flow rate and/or temperature of the low temperature cooling medium.

[0024] After the freezing of the batch to be freeze dried the temperature thereof is 0 - -60°C. The temperature of the frozen batch to be freeze dried is dependent on the solvent to be sublimated and the nature of the batch to be freeze dried.

[0025] Preferably after freezing the temperature of the batch to be freeze dried is -55 - -15°C.

[0026] A suitable vacuum for sublimating the solvent is a vacuum varying from 5 mbar - 0.01 mbar (preferably < 0.1 mbar. The one and the other are of course dependent on the solvent to be sublimated and the nature of the batch to be freeze dried.

[0027] At the application with a technically used conical mixer for a 50 litre batch, a two or more staged vacuum pumping system has been extremely suitable in connection with the process control.

[0028] Of course it is possible at freeze drying solutions, such as solutions of proteins and/or other pharmaceutical products to apply a so-called dry single vacuum pump without condenser, such as described in US-A-594488144 or the corresponding WO 99/18402.

[0029] The application of such a vacuum system is dependent on the solvent vapour which is expected to be sublimated.

[0030] It is furthermore remarked, that in column 4, lines 14 - 16 of US-A-5.948.144 is mentioned "Lyophilized cakes looked as good as or better than those produced with a cold trap condenser in operation". With this prior known method for freeze drying of solutions no free flowing product is therefore obtained, but a cake. US-A-5.948.144 namely relates to a freeze dryer of the "tray-on-cabinet" type mentioned in the introduction.

[0031] With relation to the necessary small clearance between the wall of the conical vessel 1 and the mixing member which scrapes the wall, it is remarked that this clearance is preferably between 0.5 and 15 mm.

[0032] More preferably the clearance between the wall and the mixing member is between 1 and 10 mm.

[0033] Without limitation to a certain type of wall scraping mixing member, preferred mixing members are a centrally driven ribbon shaped element, a centrally driven blade element or an orbital screw element provided with a swing arm.

[0034] As freeze drying is a slow process, characterizing drying times of 10 - 100 hours for a 50 litre batch being usual, only a small mixing effect during the freeze

drying step is necessary. The above mentioned mixing members are excellently suitable for this. The little mixing action during the freeze drying precludes a heat import that is too high and surprisingly a product of excellent quality is obtained.

[0035] The use of a certain type of element as mixing member is dependent on the necessary application, which is again dependent on the material to be freeze dried and the freezing method used.

[0036] When using a mixing member that scrapes a wall it is of predominant importance, that the member or element 3 sweeps during its movement along the complete area in which the product to be obtained is in contact with the wall 2.

[0037] As small vacuum leakage rates are essential for maintaining a high vacuum level, according to a preferred embodiment the mixing member 3 with its drive 6 is coupled by a magnetic coupling 16.

[0038] Another preferred solution is that only a drive 6 is applied for the mixing member 3 having only a rotary seal with respect to the vessel 1.

[0039] In the case in which relative low temperatures are applied, according to a preferential embodiment the drive unit 6 has been constructed in such a way, that this one has bearings 7 outside the vessel 1. This is to preclude the freezing of the lubricants in the bearings 7, by which operational breakdowns are caused.

[0040] According to a preferential embodiment of the novel method, the conical vessel 1 is connected to a condenser 8, such as known from the earlier mentioned US-A-4.245.399, viz. fig. 1, reference no. 48 and column 3, lines 58 and 59.

[0041] As when freeze drying solutions and such vary great quantities of water have to be removed there from, this condenser 8 has spacing between the cooling tubes 9 of at least 3 - 4 cm for operational conditions to a maximal ice layer of about 1 cm on the cooling tubes 9.

[0042] In connection with the great quantity of solvent (water vapour) to be sublimated, according to a preferred embodiment a double condenser 8 is used, of which one is in operation during freeze drying and the other is being defrosted.

[0043] According to a preferred embodiment a filter 10 having a great filter surface is used between the condenser 8 and the vessel 1, to preclude fouling the condenser 8 with dust from the products to be dried during freeze drying. The filter should have a great filter surface in order to keep the pressure drop over the filter at a low value.

[0044] At high vacuum levels it is preferred however to avoid even this pressure drop and make a direct connection between the condenser 8 and the vessel 1.

[0045] According to another preferred embodiment with the direct connection of the condenser 8 and the vessel 1, the condenser is arranged next to the vessel 1.

[0046] It is also possible to provide the filter 8 with a heat exchanger 16.

[0047] The invention is now elucidated in the following

specification with 9 figures, in which the same reference numerals are used for parts having similar functions.

[0048] It is remarked, that not all of the above mentioned aspects of the invention are elucidated in the enclosed figures, inter alia it has not been elucidated the direct freezing by means of liquid nitrogen or solid carbon dioxide and the use of a double condenser.

[0049] In the figures are shown:

Fig. 1 a general system survey of stirred freeze drying according to the invention;

Fig. 2 a structure in which the condenser is directly mounted above the drying vessel;

Fig. 3 a structure in which the condenser is arranged next to the drying vessel;

Fig. 4 a structure in which the filter is present between the drying vessel and the condenser;

Fig. 5 a structure in which a centrally driven ribbon shaped mixing member is used in the drying vessel;

Fig. 6 a structure in which a centrally driven blade member is used in the drying vessel;

Fig. 7 a structure in which an orbital screw element provided with a swing arm is used in the drying vessel;

Fig. 8 a structure in which the drive unit and the bearings are present outside and at a distance from the drying vessel;

Fig. 9 a structure in which a magnetic coupling is present between the drive of the mixing member outside the drying vessel and the mixing member itself inside the drying vessel.

Description of the figures

[0050] In the general scheme of the system are shown in fig. 1 the vessel 1, the inner wall 2 thereof, the mixing member 3 and the jacket 4 of the vessel 1.

[0051] In the upper side of the vessel 1 there is a condenser 8 which is connected with a tool stage vacuum system 5 that comprises two vacuum pumps 13 and 14. The vessel jacket 4 is connected to both a cooling unit 11 and a heating unit 12.

[0052] The freeze drying method is controlled with a control unit 15 for pressure and temperature, which is indicated with P and T.

[0053] In the control scheme the drive 6 of the mixing member shown with M, M or 6 has not been included in the circuit of the control unit 15. The fact that the drive 5 or 6 has not been included in this circuit is caused by the

slow and good mixing of the batch in the vessel 1 is such that including the power of the mixing engine in the circuit is unnecessary.

[0054] Nevertheless, this leads to an extremely good product and a simplification of the necessary control circuit.

Fig. 2 is a view of a structure of which the condenser 8 is directly on the vessel 1.

Fig. 3 is a view of a structure in which the condenser 8 is mounted next to the vessel 1 in order to preclude the splashing of material from the vessel onto the condenser 8.

Fig. 4 is a view of a structure in which a filter 10, comprising a heat exchanger 16 is mounted between the vessel 1 and the condenser 8. The heat exchanger 16 serves to counteract condensation of vapour by heating, which condensed vapour, could clog the filter 10.

Fig. 5 is a view of a structure in which a centrally driven ribbon shaped mixing member 3a is mounted inside the vessel 1.

Fig. 6 and 7 are view of structures in which a centrally driven blade mixing member 3b and a orbital screw mixing element comprising a swing arm 3c are in the vessel 1 respectively.

Fig. 8 is a view of a structure of which the bearings 7 of the drive unit 6 for the mixing element 3 are present outside the vessel 1.

[0055] As mentioned earlier the bearings 7 are present outside the vessel 1 in connection with precluding the freezing of the lubricants in the vessel.

[0056] Fig. 9 is a view of a structure of which the drive 6 is coupled by means of a magnetic coupling 16 with the mixing element 3. As mentioned earlier, this structure is applied in order to minimize the vacuum leak losses as much as possible. The wall 2 of the vessel 1 is in this case not punctured by a bore for a shaft.

[0057] It is obvious that within the scope of the enclosed claims alterations of the structures shown in the figures are possible.

[0058] With respect to the application of the novel method for freeze drying in can be mentioned, that the most important application lies in the area of the pharmaceutical industry. About 30% of all antibiotics, 90% of the macro molecules and 50% of the electrolytes are produced by the application of freeze drying.

[0059] Other products which are produced in a characterizing way with freeze drying are proteins, hormones, vaccines, bacteria, yeast, blood serum, liposome and transplanted materials, such as collagens sponge.

[0060] The deciding factor to apply freeze drying with all these structures is the conservation of structure and the low temperature load.

[0061] In the area of the food industry lies a second important area of use of freeze drying. Maintaining taste, improved storage life and ready to use characteristics

are important questions. Vegetables, potatoes, fruit, juices, coffee, eggs and rice are characterizing freeze dried materials.

[0062] A third important area of use for freeze drying on a large scale lies in the area of new materials. More particularly one can think for this to metal oxide, ceramics, special composite materials and nano materials. Applications of freeze drying with these materials have special advantages. During the freezing and drying the particles suspended in a liquid or solvent remain separated, because of which no hard caked together agglomerates are obtained. It is even possible to obtain homogeneous mixtures at a molecular scale by freeze drying. This is an important completely new development, in which new mixtures of two or more substances are obtained at a molecular scale.

[0063] The novel homogeneous mixtures of two or more substances on molecular scale are extremely well useable in the industry of novel materials.

Claims

1. Method for drying solutions and liquid containing solid substances, in which a vessel (1) having a downwardly conical shape is applied having inside a mixing member (3) which rotates along the wall (2) of the vessel, said mixing member (3) rotates with a small interspace along said wall (2), whereas the vessel (1) is provided with a jacket (4) and is connected with a vacuum system (5), wherein one freeze dries a batch inside said conical vessel (1), **characterized in that** inside said is applied to the batch.
2. Method according to claim 1, **characterized in that**, said conical vessel (1) is a part of a conical mixer (6).
3. Method according to claim 1 or 2, **characterized in that**, a batch to be freeze dried in said conical vessel (1) is frozen by direct cooling.
4. Method according to claim 3, **characterized in that**, one introduces liquid nitrogen or solid carbon dioxide into said conical vessel (1) and mixes that with the batch to be freeze dried.
5. Method according to one or more of the preceding claims, **characterized in that**, one freezes the batch to be cooled with a freezing rate of 0.1 - 10°C/minute.
6. Method according to claim 5, **characterized in that**, after freezing the temperature of the batch to be freeze dried in the vessel (1) is 0 - -60°C.
7. Method according to claim 6, **characterized in that**, after freezing the temperature of the batch to be freeze dried is -55 - -15°C.
8. Method according to one or more of the preceding claims, **characterized in that**, during the vacuum drying the vacuum varies from 5 mbar - 0.01 mbar, preferably < 0.1 mbar.
9. Method according to claim 8, **characterized in that**, for maintaining the vacuum one uses a two or more stage vacuum pump system (5).
10. Method according to one or more of the preceding claims, **characterized in that**, inside the conical vessel (1) a wall (2) scraping mixing member (3) is applied with a small inter-space between the wall (2) and the mixing member (3), said inter-space amounting to 0.5 - 1.5 mm.
11. Method according to one or more of the preceding claims 1 to 9, **characterized in that**, inside the conical vessel (1) a wall (2) scraping mixing member (3') is applied with a small inter-space between the wall (2) and the mixing member (3), said inter-space amounting to 1 - 10 mm.
12. Method according to one or more of the preceding claims, **characterized in that**, one uses as mixing member (3) a centrally driven ribbon shaped element, a centrally driven blade element or an orbital screw element provided with a swing arm.
13. Method according to one or more of the preceding claims, **characterized in that**, the mixing member (3) is coupled with its drive (6) through a magnetic coupling (16).
14. Method according to each of the claims 1 - 12, **characterized in that**, only one drive (6) is used for the mixing member (3) with only a rotary seal with respect to the vessel (1).
15. Method according to each of the preceding claims, **characterized in that**, the drive (6) has bearings (7) outside the vessel (1).
16. Method according to one or more of the preceding claims, in which said conical vessel (1) is connected with a condenser (8), **characterized in that**, the condenser (8) has an inter-space between the cooling tubes (9) of at least 3 - 4 cm, for operating conditions until a maximum ice-layer of about 1 cm on the cooling tubes (9).
17. Method according to claim 16, **characterized in that**, one uses a double condenser (8).
18. Method according to one or more of the preceding claims, **characterized in that**, one uses between the condenser (8) and the vessel (1) a filter (10) having a large filter surface.

19. Method according to claims 16, **characterized in that**, the condenser (8) is directly connected with the vessel (1).
20. Method according to claims 16 or 19, **characterized in that**, the condenser (8) is mounted next to the vessel (1).
21. Device for using the method according to one or more of the claims 1 - 20, comprising a vessel having a downwardly conical shape (1), with a mixing member (3) which rotates along the wall (2) of the vessel (1), in which said mixing member (3) rotates with a small inter-space along the wall (2) of the vessel, said vessel (1) being provided with a jacket (4) and being connected with a vacuum system (5), wherein, said conical vessel (1) is arranged for freeze drying solutions and liquid containing solid substances, **characterized in that** the device is arranged for applying a freeseing step and subsequently a drying step to a batch inside said conical vessel (1).
22. Device according to claim 21, **characterized in that**, said conical vessel (1) being part of a conical mixer.
23. Device according to claim 21 or 22, **characterized in that**, the device further comprises a cooling unit (11) and a heating unit (12) of the jacket (4) and a condenser (8) placed behind the vessel (1), in which the condenser (8) is connected to a two-stage vacuum pump system (5) which comprises two vacuum pumps (13) and (14) and a control unit (15) for the pressure and temperature in the vessel (1).
24. Device according to one of the claims 21 - 23, **characterized in that**, the condenser (8) is placed above the vessel (1).
25. Device according to one of the claims 21 - 23, **characterized in that**, the condenser (8) is placed next to the vessel (1).
26. Device according to claim 24 or 25, **characterized in that**, a filter (10) is placed between the vessel (1) and the condenser (8).
27. Device according to claim 26, **characterized in that**, the filter (10) is provided with a heat exchanger (16).
28. Device according to one or more of the claims 21 - 27, **characterized in that**, inside the vessel (1) a centrally driven ribbon shaped mixing member (3a) is present.
29. Device according to one or more of the claims 21 - 27, **characterized in that**, inside the vessel (1) a centrally driven blade mixing member (3b) is present.

30. Device according to each of the claims 21 - 27, **characterized in that**, inside the vessel (1) an orbital screw element or mixing member with a swing arm (3c) is present.
31. Device according to one or more of the claims 21 - 30, **characterized in that**, the drive unit (6) and the corresponding bearing (7) are present outside the vessel (1).
32. Device according to each of the claims 21 - 30, **characterized in that**, the mixing member (3) is coupled with its drive (6) by means of a magnetic coupling (16).
33. Device according to one or more of the claims 21 - 32, **characterized in that**, the condenser (8) being double.

Patentansprüche

- Verfahren zum Trocknen von Lösungen und von Flüssigkeit enthaltenden festen Substanzen, wobei ein Gefäß (1) mit einer nach unten hin konischen Gestalt eingesetzt wird, welches innen ein Mischelement (3) aufweist, welches entlang der Wand (2) des Gefäßes rotiert, wobei das Mischelement (3) mit einem kleinen Abstand entlang der Wand (2) rotiert, und wobei das Gefäß (1) mit einem Mantel (4) versehen ist und mit einem Vakuumsystem (5) verbunden ist, bei dem man eine Charge innerhalb des konischen Gefäßes (1) gefriertrocknet, **dadurch gekennzeichnet, dass** in dem konischen Gefäß (1) auf die Charge ein Gefrierschritt und anschließend ein Trocknungsschritt angewandt wird.
- Verfahren nach Anspruch 1, **dadurch gekennzeichnet, dass** das konische Gefäß (1) ein Teil eines konischen Mischers (6) ist.
- Verfahren nach Anspruch 1 oder 2, **dadurch gekennzeichnet, dass** eine in dem konischen Gefäß (1) zu gefriertrocknende Charge durch direkte Kühlung eingefroren wird.
- Verfahren nach Anspruch 3, **dadurch gekennzeichnet, dass** man flüssigen Stickstoff oder festes Kohlenstoffdioxid in das konische Gefäß (1) einführt und diesen/s mit der zu gefriertrocknenden Charge mischt.
- Verfahren nach einem oder mehreren der vorhergehenden Ansprüche, **dadurch gekennzeichnet, dass** man die zu kühlende Charge mit einer Gefrier-rate von 0,1 - 10°C/Minute einfriert.

6. Verfahren nach Anspruch 5, **dadurch gekennzeichnet, dass** nach dem Einfrieren die Temperatur der in dem Gefäß (1) zu gefriertrocknenden Charge 0 bis -60°C beträgt.
7. Verfahren nach Anspruch 6, **dadurch gekennzeichnet, dass** nach dem Einfrieren die Temperatur der zu gefriertrocknenden Charge -55 bis -15°C beträgt.
8. Verfahren nach einem oder mehreren der vorhergehenden Ansprüche, **dadurch gekennzeichnet, dass** während des Vakuumtrocknens das Vakuum von 5 mbar - 0,01 mbar variiert, bevorzugt < 0,1 mbar.
9. Verfahren nach Anspruch 8, **dadurch gekennzeichnet, dass** man zum Aufrechterhalten des Vakuums ein zwei- oder mehrstufiges Vakuumpumpensystem (5) verwendet.
10. Verfahren nach einem oder mehreren der vorhergehenden Ansprüche, **dadurch gekennzeichnet, dass** in dem konischen Gefäß (1) ein Wand(2)-Abstreichmischelement (3) mit einem kleinen Abstand zwischen der Wand (2) und dem Mischelement (3) verwendet wird, wobei der Abstand 0,5 - 1,5 mm beträgt.
11. Verfahren nach einem oder mehreren der vorhergehenden Ansprüche 1-9, **dadurch gekennzeichnet, dass** in dem konischen Gefäß (1) ein Wand(2)-Abstreichmischelement (3) mit einem kleinen Abstand zwischen der Wand (2) und dem Mischelement (3) verwendet wird, wobei der Abstand 1 - 10 mm beträgt.
12. Verfahren nach einem oder mehreren der vorhergehenden Ansprüche, **dadurch gekennzeichnet, dass** man als Mischelement (3) ein mittig angetriebenes bandförmiges Element, ein mittig angetriebenes Klingenelement oder ein umlaufendes Schraubenelement, das mit einem Schwingarm versehen ist, verwendet.
13. Verfahren nach einem oder mehreren der vorhergehenden Ansprüche, **dadurch gekennzeichnet, dass** das Mischelement (3) mittels einer Magnetkupplung (16) mit seinem Antrieb (6) verbunden ist.
14. Verfahren nach jedem der Ansprüche 1 - 12, **dadurch gekennzeichnet, dass** nur ein Antrieb (6) für das Mischelement (3) mit nur einer Rotationsdichtung bezüglich des Gefäßes (1) verwendet wird.
15. Verfahren nach jedem der vorhergehenden Ansprüche, **dadurch gekennzeichnet, dass** der Antrieb (6) Lager (7) außerhalb des Gefäßes (1) aufweist.
16. Verfahren nach einem oder mehreren der vorhergehenden Ansprüche, wobei das konische Gefäß (1) mit einem Kondensator (8) verbunden ist, **dadurch gekennzeichnet, dass** der Kondensator (8) einen Abstand von mindestens 3 - 4 cm zwischen den Kühlrohren (9) aufweist, damit Betriebsbedingungen bis zu einer maximalen Eisschicht von etwa 1 cm auf den Kühlrohren (9) möglich sind.
17. Verfahren nach Anspruch 16, **dadurch gekennzeichnet, dass** man einen doppelten Kondensator (8) verwendet.
18. Verfahren nach einem oder mehreren der vorhergehenden Ansprüche, **dadurch gekennzeichnet, dass** man zwischen dem Kondensator (8) und dem Gefäß (1) einen Filter (10) mit einer großen Filteroberfläche verwendet.
19. Verfahren nach Anspruch 16, **dadurch gekennzeichnet, dass** der Kondensator (8) direkt mit dem Gefäß (1) verbunden ist.
20. Verfahren nach Anspruch 16 oder 19, **dadurch gekennzeichnet, dass** der Kondensator neben dem Gefäß (1) angebracht ist.
21. Vorrichtung zur Verwendung des Verfahrens nach einem oder mehreren der Ansprüche 1 - 20, umfassend ein Gefäß mit nach unten hin konischer Gestalt, welches ein Mischelement (3) aufweist, welches entlang der Wand (2) des Gefäßes rotiert, wobei das Mischelement (3) mit einem kleinen Abstand entlang der Wand (2) des Gefäßes rotiert, wobei das Gefäß (1) mit einem Mantel (4) versehen und mit einem Vakuumsystem (5) verbunden ist, wobei das konische Gefäß (1) zum Gefriertrocknen von Lösungen und von Flüssigkeit enthaltenden festen Substanzen ausgelegt ist, **dadurch gekennzeichnet, dass** die Vorrichtung dafür ausgelegt ist, auf eine Charge in dem konischen Gefäß (1) einen Gefrierschritt und anschließend einen Trocknungsschritt anzuwenden.
22. Vorrichtung nach Anspruch 21, **dadurch gekennzeichnet, dass** das konische Gefäß (1) Teil eines konischen Mischers ist.
23. Vorrichtung nach Anspruch 21 oder 22, **dadurch gekennzeichnet, dass** die Vorrichtung weiterhin eine Kühleinheit (11) und eine Heizeinheit (12) des Mantels (4) und einen hinter dem Gefäß (1) angeordneten Kondensator (8) aufweist, wobei der Kondensator (8) an ein zweistufiges Vakuumpumpensystem (5) angeschlossen ist, welches zwei Vakuumpumpen (13) und (14) umfasst, sowie eine Steuerungseinheit (15) für den Druck und die Temperatur in dem

Gefäß (1) aufweist.

24. Vorrichtung nach einem der Ansprüche 21 - 23, **dadurch gekennzeichnet, dass** der Kondensator (8) über dem Gefäß (1) angeordnet ist.

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25. Vorrichtung nach einem der Ansprüche 21 - 23, **dadurch gekennzeichnet, dass** der Kondensator (8) neben dem Gefäß (1) angeordnet ist.

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26. Vorrichtung nach Anspruch 24 oder 25, **dadurch gekennzeichnet, dass** ein Filter (10) zwischen dem Gefäß (1) und dem Kondensator (8) angeordnet ist.

27. Vorrichtung nach Anspruch 26, **dadurch gekennzeichnet, dass** der Filter (10) mit einem Wärmetauscher (16) versehen ist.

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28. Vorrichtung nach einem oder mehreren der Ansprüche 21 - 27, **dadurch gekennzeichnet, dass** sich in dem Gefäß (1) ein mittig angetriebenes, bandförmiges Mischelement (3a) befindet.

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29. Vorrichtung nach einem oder mehreren der Ansprüche 21 - 27, **dadurch gekennzeichnet, dass** sich in dem Gefäß (1) ein mittig angetriebenes Klingen-Mischelement (3b) befindet.

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30. Vorrichtung nach jedem der Ansprüche 21 - 27, **dadurch gekennzeichnet, dass** sich in dem Gefäß (1) ein umlaufendes Schraubenelement oder Mischelement mit einem Schwingarm (3c) befindet.

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31. Vorrichtung nach einem oder mehreren der Ansprüche 21 - 30, **dadurch gekennzeichnet, dass** sich die Antriebseinheit (6) und das entsprechenden Lager (7) außerhalb des Gefäßes (1) befinden.

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32. Vorrichtung nach jedem der Ansprüche 21 - 30, **dadurch gekennzeichnet, dass** das Mischelement (3) mit seinem Antrieb (6) mittels einer Magnetkupplung (16) verbunden ist.

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33. Vorrichtung nach einem oder mehreren der Ansprüche 21 - 32, **dadurch gekennzeichnet, dass** der Kondensator (8) zweifach vorliegt.

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Revendications

1. Procédé de séchage de solutions et de liquide contenant des substances solides, dans lequel une cuve (1) ayant une forme conique vers le bas est appliquée comportant à l'intérieur un organe de mélange (3) qui tourne le long de la paroi (2) de la cuve, ledit organe de mélange (3) tourne avec un petit espacement le long de ladite paroi (2), tandis que la cuve (1) est pourvue d'une chemise (4) et est raccordée

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- à un système de vide (5), dans lequel on lyophilise un lot à l'intérieur de ladite cuve conique (1), **caractérisé en ce qu'à** l'intérieur de ladite cuve conique (1), une étape de congélation et ultérieurement une étape de séchage sont appliquées au lot.

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2. Procédé selon la revendication 1, **caractérisé en ce que** ladite cuve conique (1) fait partie d'un mélangeur conique (6).

3. Procédé selon la revendication 1 ou 2, **caractérisé en ce qu'un** lot à lyophiliser dans ladite cuve conique (1) est congelé par refroidissement direct.

4. Procédé selon la revendication 3, **caractérisé en ce que** l'on introduit de l'azote liquide ou du dioxyde de carbone solide dans ladite cuve conique (1) et on le mélange avec le lot à lyophiliser.

5. Procédé selon une ou plusieurs des revendications précédentes, **caractérisé en ce que** l'on congèle le lot à refroidir avec une vitesse de congélation de 0,1 à 10 °C/minute.

6. Procédé selon la revendication 5, **caractérisé en ce que**, après congélation, la température du lot à lyophiliser dans la cuve (1) est de 0 à -60 °C.

7. Procédé selon la revendication 6, **caractérisé en ce que**, après congélation, la température du lot à lyophiliser est de -55 à -15 °C.

8. Procédé selon une ou plusieurs des revendications précédentes, **caractérisé en ce que** pendant le séchage sous vide le vide varie de 5 mbar à 0,01 mbar, de préférence < 0,1 mbar.

9. Procédé selon la revendication 8, **caractérisé en ce que**, pour maintenir le vide, on utilise un système de pompe à vide à deux étages ou plus (5).

10. Procédé selon une ou plusieurs des revendications précédentes, **caractérisé en ce que**, à l'intérieur de la cuve conique (1), un organe de mélange et raclage (3) de paroi (2) est appliqué avec un petit espacement entre la paroi (2) et l'organe de mélange (3), ledit espacement s'élevant à 0,5 à 1,5 mm.

11. Procédé selon une ou plusieurs des revendications 1 à 9 précédentes, **caractérisé en ce que**, à l'intérieur de la cuve conique (1), un organe de mélange et raclage (3) de paroi (2) est appliqué avec un petit espacement entre la paroi (2) et l'organe de mélange (3), ledit espacement s'élevant à 1 à 10 mm.

12. Procédé selon une ou plusieurs des revendications précédentes, **caractérisé en ce que** l'on utilise comme organe de mélange (3) un élément en forme de

ruban entraîné centralement, un élément de pale entraîné centralement ou un élément de vis orbitale pourvu d'un bras oscillant.

13. Procédé selon une ou plusieurs des revendications précédentes, **caractérisé en ce que** l'organe de mélange (3) est couplé avec son entraînement (6) par l'intermédiaire d'un couplage magnétique (16).
14. Procédé selon chacune des revendications 1 à 12, **caractérisé en ce que** seul un entraînement (6) est utilisé pour l'organe de mélange (3) avec seulement un joint rotatif par rapport à la cuve (1).
15. Procédé selon chacune des revendications précédentes, **caractérisé en ce que** l'entraînement (6) comporte des paliers (7) à l'extérieur de la cuve (1).
16. Procédé selon une ou plusieurs des revendications précédentes, dans lequel ladite cuve conique (1) est raccordée avec un condenseur (8), **caractérisé en ce que** le condenseur (8) a un espacement entre les tubes de refroidissement (9) d'au moins 3 à 4 cm, pour des conditions de fonctionnement jusqu'à un maximum de couche de glace d'environ 1 cm sur les tubes de refroidissement (9).
17. Procédé selon la revendication 16, **caractérisé en ce que** l'on utilise un condenseur double (8).
18. Procédé selon une ou plusieurs des revendications précédentes, **caractérisé en ce que** l'on utilise entre le condenseur (8) et la cuve (1), un filtre (10) ayant une grande surface de filtre.
19. Procédé selon la revendication 16, **caractérisé en ce que** le condenseur (8) est directement raccordé à la cuve (1).
20. Procédé selon les revendications 16 ou 19, **caractérisé en ce que** le condenseur (8) est monté à côté de la cuve (1).
21. Dispositif permettant d'utiliser le procédé selon une ou plusieurs des revendications 1 à 20, comprenant une cuve ayant une forme conique vers le bas (1), avec un organe de mélange (3) qui tourne le long de la paroi (2) de la cuve (1), dans lequel ledit organe de mélange (3) tourne avec un petit espacement le long de la paroi (2) de la cuve, ladite cuve (1) étant pourvue d'une chemise (4) et étant raccordée à un système à vide (5), dans lequel ladite cuve conique (1) est agencée pour lyophiliser des solutions et des substances solides contenant des liquides, **caractérisé en ce que** le dispositif est agencé pour appliquer une étape de congélation et ultérieurement une étape de séchage à un lot à l'intérieur de ladite cuve conique (1).

22. Dispositif selon la revendication 21, **caractérisé en ce que** ladite cuve conique (1) fait partie d'un mélangeur conique.

23. Dispositif selon la revendication 21 ou 22, **caractérisé en ce que** le dispositif comprend en outre une unité de refroidissement (11) et une unité de chauffage (12) de la chemise (4) et un condenseur (8) placé derrière la cuve (1), dans lequel le condenseur (8) est raccordé à un système de pompe à vide à deux étages (5) qui comprend deux pompes à vide (13) et (14) et une unité de commande (15) pour la pression et la température dans la cuve (1).
24. Dispositif selon l'une des revendications 21 à 23, **caractérisé en ce que** le condenseur (8) est placé au-dessus de la cuve (1).
25. Dispositif selon l'une des revendications 21 à 23, **caractérisé en ce que** le condenseur (8) est placé au côté de la cuve (1).
26. Dispositif selon la revendication 24 ou 25, **caractérisé en ce qu'un** filtre (10) est placé entre la cuve (1) et le condenseur (8).
27. Dispositif selon la revendication 26, **caractérisé en ce que** le filtre (10) est pourvu d'un échangeur de chaleur (16).
28. Dispositif selon une ou plusieurs des revendications 21 à 27, **caractérisé en ce qu'à** l'intérieur de la cuve (1), un organe de mélange en forme de ruban entraîné centralement (3a) est présent.
29. Dispositif selon une ou plusieurs des revendications 21 à 27, **caractérisé en ce qu'à** l'intérieur de la cuve (1), un organe de mélange à pale entraîné centralement (3b) est présent.
30. Dispositif selon chacune des revendications 21 à 27, **caractérisé en ce qu'à** l'intérieur de la cuve (1), un élément à vis orbitale ou un organe de mélange avec un bras oscillant (3c) est présent.
31. Dispositif selon une ou plusieurs des revendications 21 à 30, **caractérisé en ce que** l'unité d'entraînement (6) et le palier correspondant (7) sont présents à l'extérieur de la cuve (1).
32. Dispositif selon chacune des revendications 21 à 30, **caractérisé en ce que** l'organe de mélange (3) est couplé avec son entraînement (6) au moyen d'un couplage magnétique (16).
33. Dispositif selon une ou plusieurs des revendications 21 à 32, **caractérisé en ce que** le condenseur (8) est double.

Fig.1

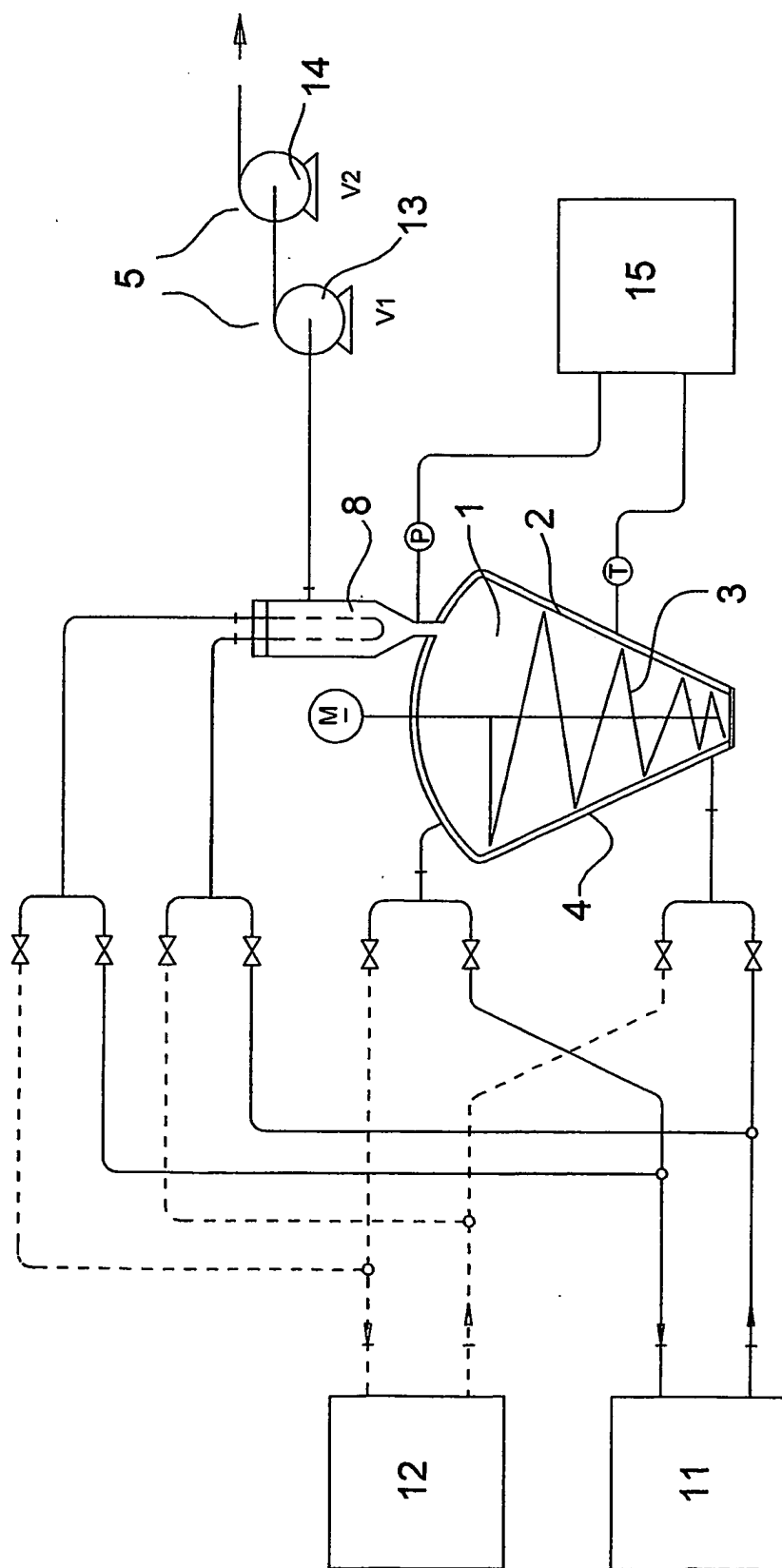


Fig.2

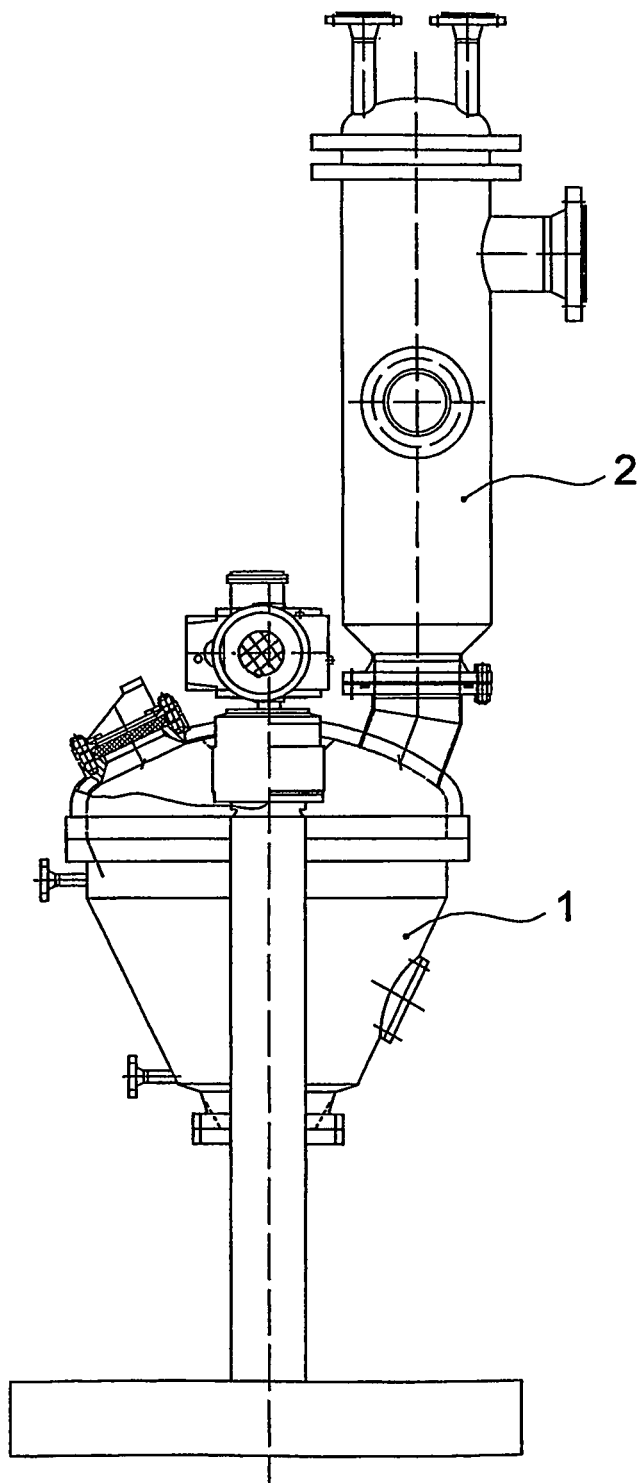


Fig.3

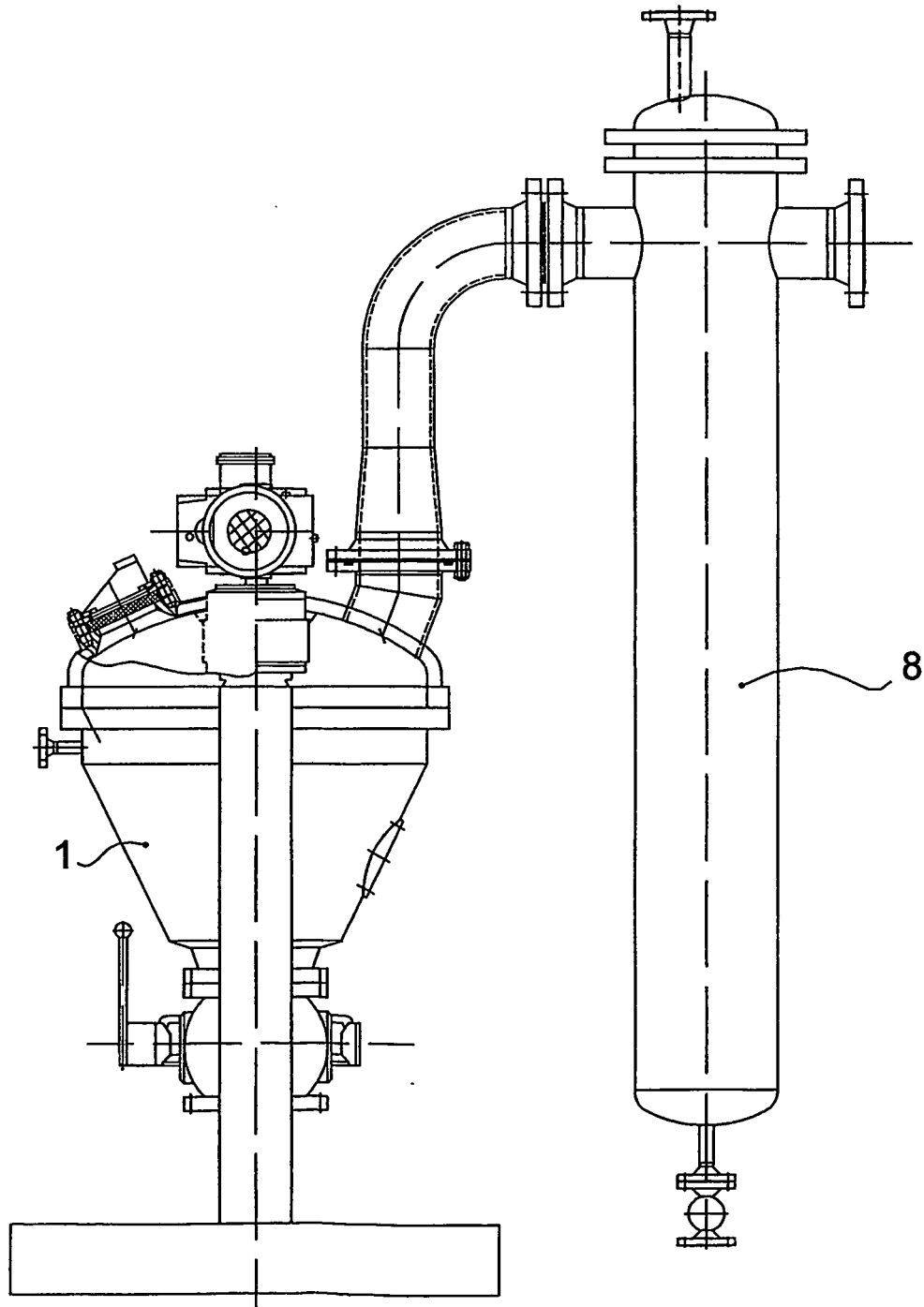


Fig.4

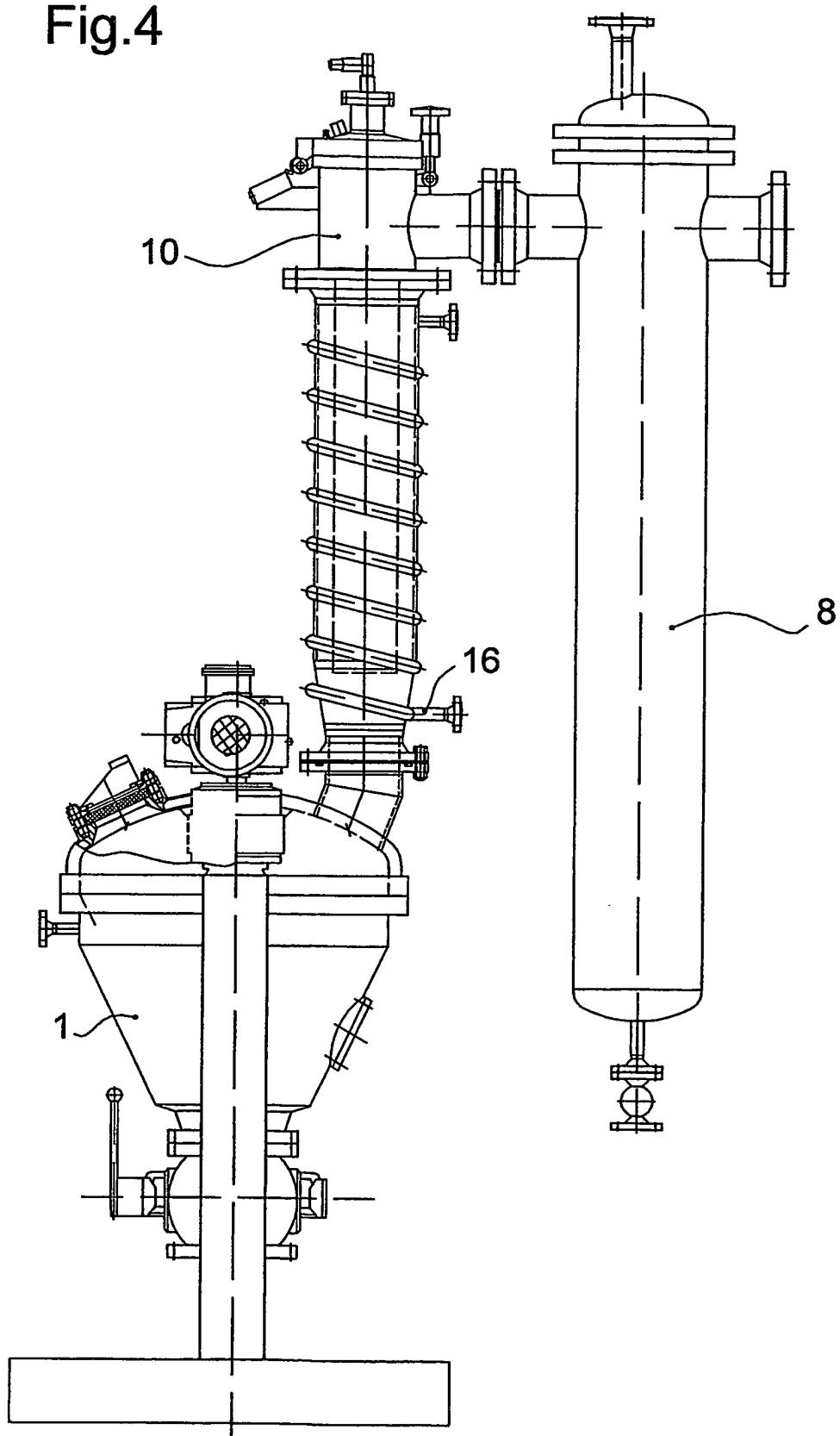


Fig.5

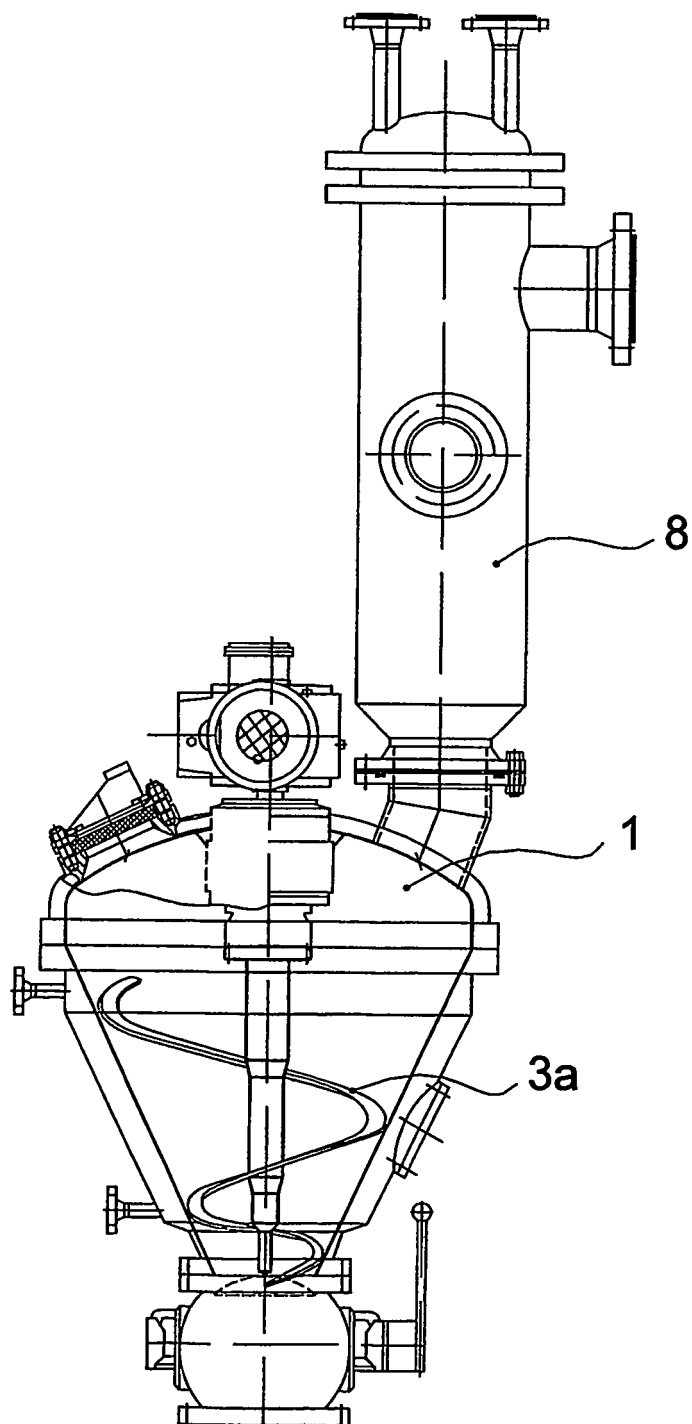


Fig.6

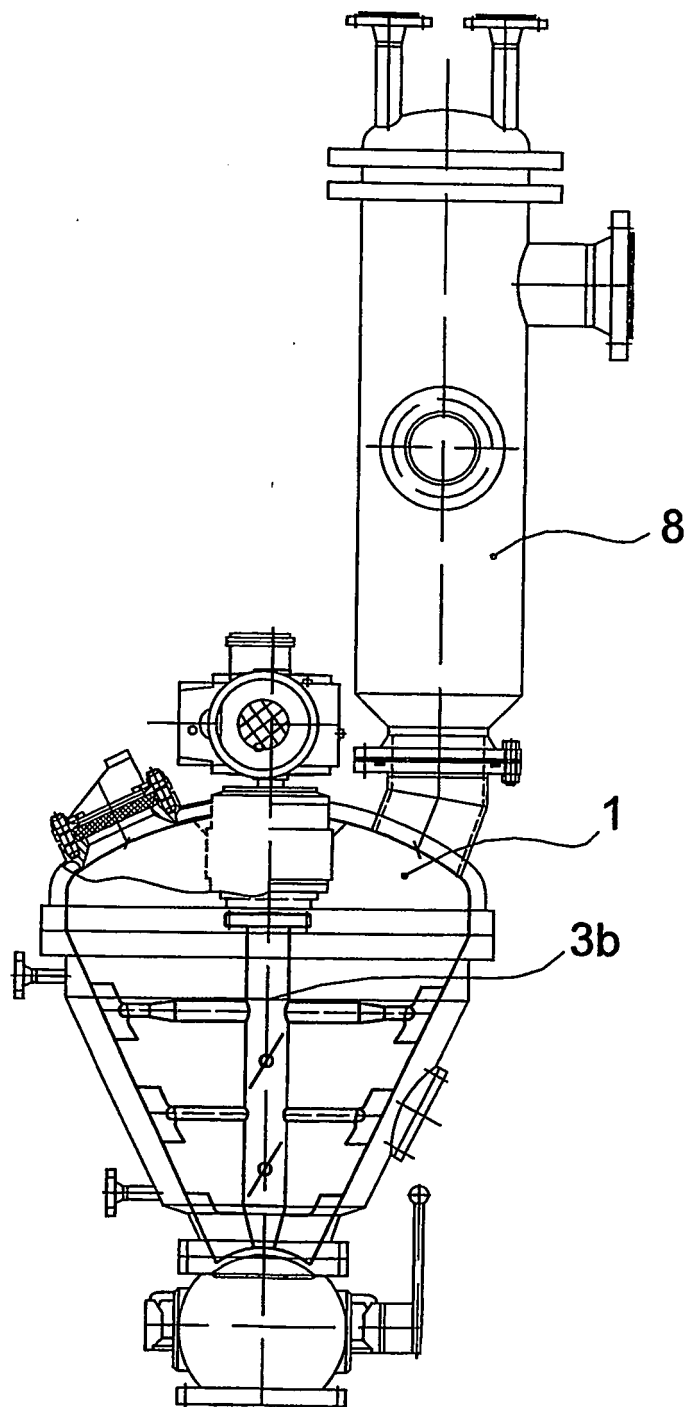


Fig.7

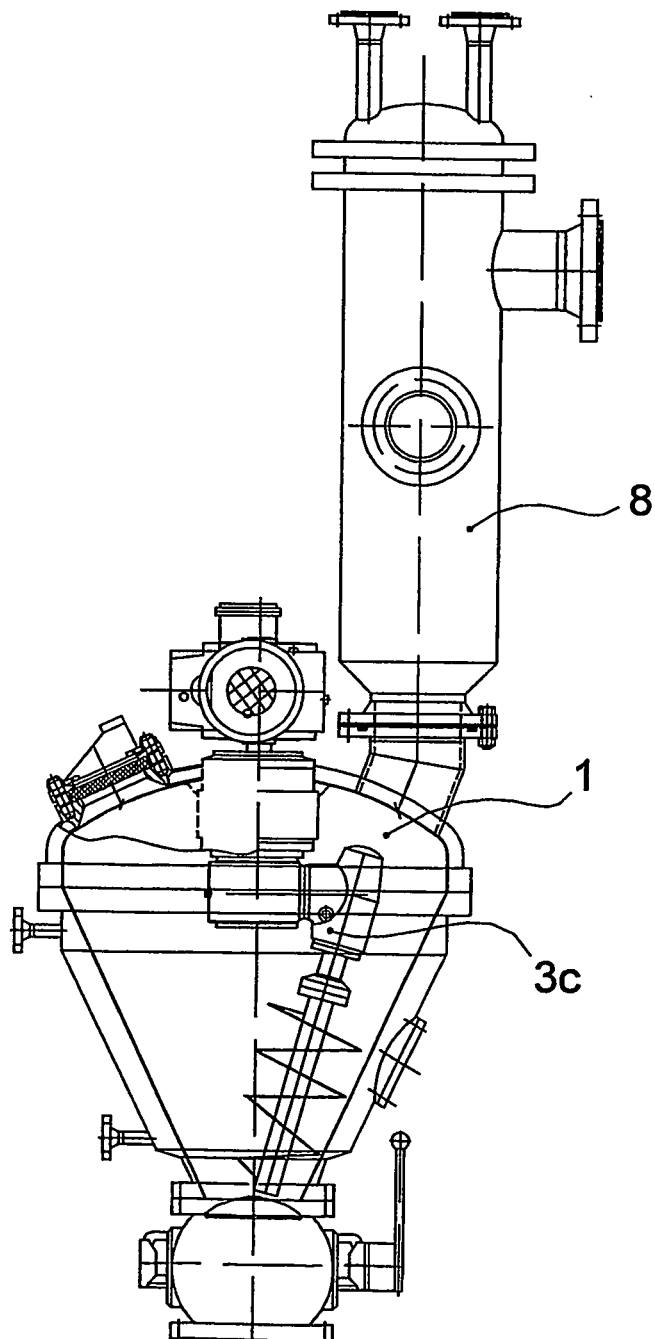


Fig.8

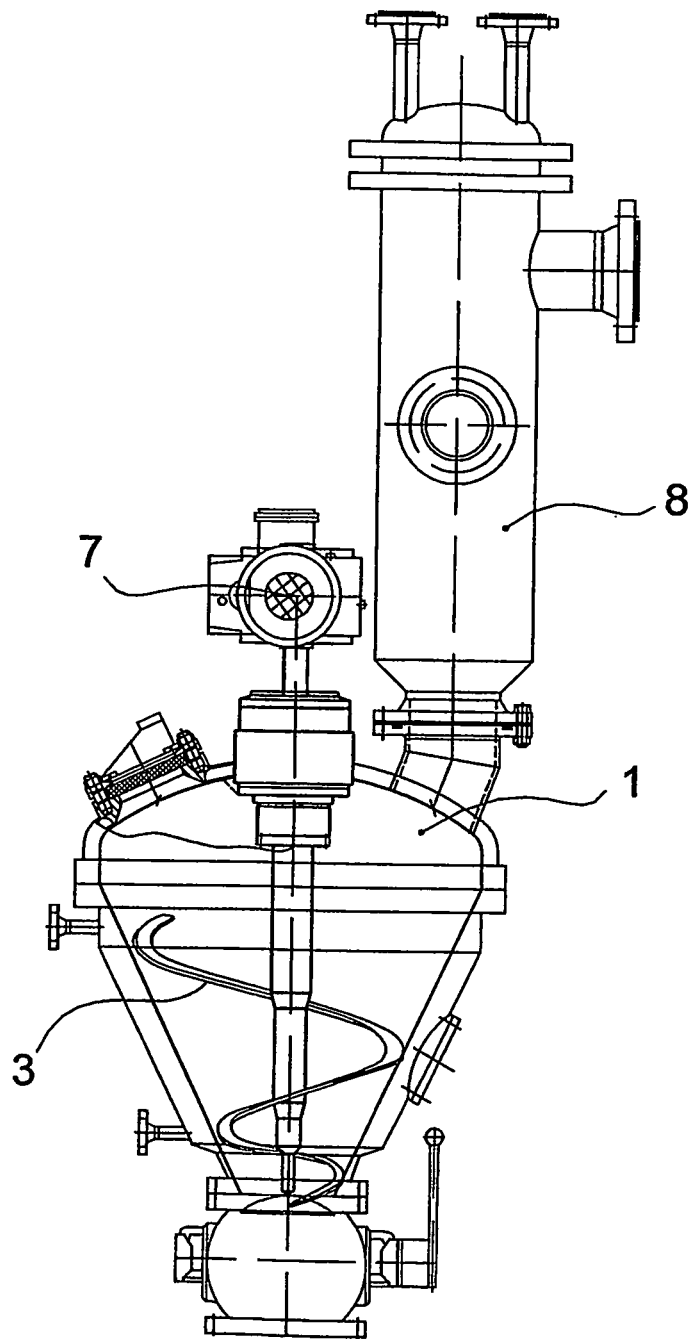
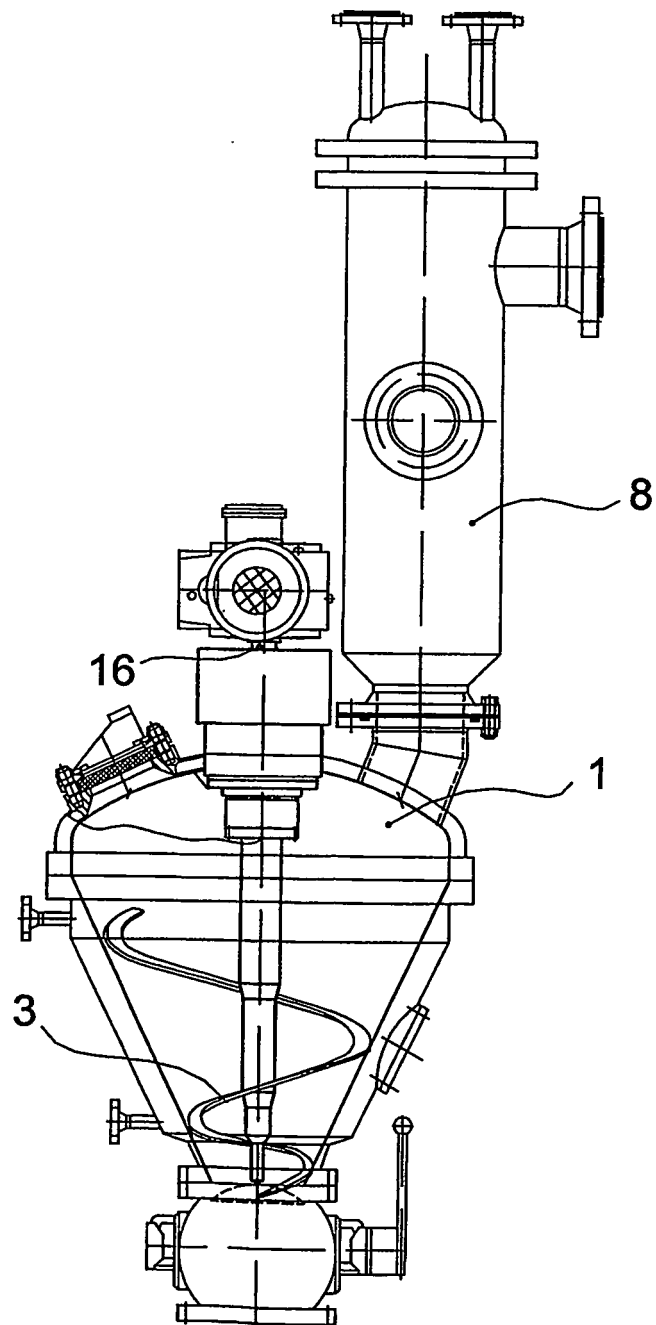


Fig.9



REFERENCES CITED IN THE DESCRIPTION

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